第八章习题答案

8-1　设半波天线上的电流分布为：$I = I_m \cos\beta z,\quad -l/2 < z < l/2$

（1）求证：当　$r_0 >> l$ 时，矢量位 $A = \vec{e_z}\dfrac{\mu_0 I_m e^{-j\beta r_0}}{2\pi\beta r_0}\dfrac{\cos\left(\dfrac{\pi}{2}\cos\theta\right)}{\sin^2\theta}$

（2）求远区的 $\left(r_0 >> l\right)$ 的磁场和电场，以及平均坡印廷矢量 S_{av}。

证明：（1）　$\boldsymbol{A} = \boldsymbol{e}_z \dfrac{\mu_0}{4\pi}\displaystyle\int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{I_{\mathrm{m}}\cos\beta z' \cdot \mathrm{d}z'}{R}\mathrm{e}^{-\mathrm{j}\beta R}$

$$R = r - z'\cos\theta$$

$$\therefore\ \boldsymbol{A} = \boldsymbol{e}_z \frac{\mu_0}{4\pi}\int_{-\frac{l}{2}}^{\frac{l}{2}} \frac{I_{\mathrm{m}}\cos\beta z'}{r}\mathrm{e}^{-\mathrm{j}\beta(r-\mathrm{z}'\cos\theta)} \cdot \mathrm{d}z'$$

$$= \boldsymbol{e}_z \frac{\mu_0 I_{\mathrm{m}}\mathrm{e}^{-\mathrm{j}\beta r}}{4\pi r}\int_{-\frac{l}{2}}^{\frac{l}{2}} \cos\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta} \cdot \mathrm{d}z'$$

用分部积分法

$$\int_{-\frac{l}{2}}^{\frac{l}{2}} \cos\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta} \cdot \mathrm{d}z' = \frac{1}{\mathrm{j}\beta\cos\theta}\cos\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta}\Big|_{-\frac{l}{2}}^{\frac{l}{2}} + \frac{\beta}{\mathrm{j}\beta\cos\theta}\int_{-\frac{l}{2}}^{\frac{l}{2}} \sin\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta} \cdot \mathrm{d}z'$$

$$= \frac{\beta}{(\mathrm{j}\beta\cos\theta)^2}\sin\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta}\Big|_{-\frac{l}{2}}^{\frac{l}{2}} - \frac{\beta^2}{(\mathrm{j}\beta\cos\theta)^2}\int_{-\frac{l}{2}}^{\frac{l}{2}} \cos\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta} \cdot \mathrm{d}z'$$

$$= \frac{\beta}{(\mathrm{j}\beta\cos\theta)^2}\sin\left(\beta\frac{l}{2}\right)\left(\mathrm{e}^{\mathrm{j}\beta\frac{l}{2}\cos\theta} + \mathrm{e}^{-\mathrm{j}\beta\frac{l}{2}\cos\theta}\right) - \frac{\beta^2}{(\mathrm{j}\beta\cos\theta)^2}\int_{-\frac{l}{2}}^{\frac{l}{2}} \cos\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta} \cdot \mathrm{d}z'$$

$$\left(1 + \frac{\beta^2}{(\mathrm{j}\beta\cos\theta)^2}\right)\int_{-\frac{l}{2}}^{\frac{l}{2}} \cos\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta} \cdot \mathrm{d}z' = \frac{2\beta}{(\mathrm{j}\beta\cos\theta)^2}\sin\left(\beta\frac{l}{2}\right)\cos\left(\beta\frac{l}{2}\cos\theta\right)$$

$$\int_{-\frac{l}{2}}^{\frac{l}{2}} \cos\beta z'\mathrm{e}^{\mathrm{j}\beta \mathrm{z}'\cos\theta} \cdot \mathrm{d}z' = \frac{2\beta}{(\mathrm{j}\beta\cos\theta)^2 + \beta^2}\sin\left(\beta\frac{l}{2}\right)\cos\left(\beta\frac{l}{2}\cos\theta\right)$$

$$\beta=\frac{2\pi}{\lambda}\qquad 2l=\lambda/2$$

$$\sin\left(\beta\frac{l}{2}\right)=\sin\frac{\pi}{2}\qquad\qquad \cos\left(\frac{\pi}{2}\cos\theta\right)$$

$$\frac{2\beta}{(\mathrm{j}\beta\cos\theta)^2+\beta^2}=\frac{2}{\beta\sin^2\theta}$$

$$\therefore\quad \int_{-\frac{l}{2}}^{\frac{l}{2}}\cos\beta z'\mathrm{e}^{\mathrm{j}\beta z'\cos\theta}\cdot\mathrm{d}z'=\frac{2}{\beta\sin^2\theta}\cos\left(\frac{\pi}{2}\cos\theta\right)$$

$$\therefore\ \boldsymbol{A}=\boldsymbol{e}_z\frac{\mu_0 I_\mathrm{m}\mathrm{e}^{-\mathrm{j}\beta r}}{4\pi r}\int_{-\frac{l}{2}}^{\frac{l}{2}}\cos\beta z'\mathrm{e}^{\mathrm{j}\beta z'\cos\theta}\cdot\mathrm{d}z'$$

$$=\boldsymbol{e}_z\frac{\mu_0 I_\mathrm{m}\mathrm{e}^{-\mathrm{j}\beta r}}{2\pi\beta r}\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin^2\theta}$$

（2）

$$\dot{E}_\theta=\ \mathrm{j}\frac{60I_\mathrm{m}}{r}\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\ \mathrm{e}^{-\mathrm{j}\beta r}$$

$$\dot{H}_\phi=\ \mathrm{j}\frac{I_\mathrm{m}}{2\pi r}\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\ \mathrm{e}^{-\mathrm{j}\beta r}$$

$$\boldsymbol{S}_{av}=\frac{1}{2}\mathrm{Re}\left[\dot{\boldsymbol{E}}\times\boldsymbol{H}^*\right]=\frac{1}{2\times120\pi}\left|\dot{E}_\theta\right|^2$$

$$=\frac{1}{2\times120\pi}\left[\frac{60I_\mathrm{m}}{r}\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\right]^2=\frac{15I_\mathrm{m}}{\pi r^2}\frac{\cos^2(\frac{\pi}{2}\cos\theta)}{\sin^2\theta}$$

8-2　一长为$20m$的发射天线，在频率$f=1\,\mathrm{M\,Hz}$时，可视为电偶极子天线，设天线上电流振幅的有效值为$2.5\mu\mathrm{A}$，求天线的辐射电阻R_{rad}和辐射功率P，如频率变为$f=100\,\mathrm{k\,Hz}$，则辐射功率和辐射电阻又为多少?

解：当$f=1\,\mathrm{M\,Hz}$时

$$\lambda=\frac{c}{f}=\frac{3\times10^{8}}{1\times10^{6}}=300m\quad,\qquad R_{rad}=80\pi^{2}(\frac{\Delta l}{\lambda})^{2}=3.5(\Omega$$

$$P=I^{2}R_{rad}=(2.5\times10^{-6})^{2}\times3.5=2.19\times10^{-11}\ (\mathrm{W})$$

当$f=100\,\mathrm{M\,Hz}$时

$$\lambda=\frac{c}{f}=\frac{3\times10^{8}}{100\times10^{3}}=3000m\ ,\qquad R_{rad}=80\pi^{2}(\frac{\Delta l}{\lambda})^{2}=3.5\times10^{-2}(\Omega$$

$$P=I^{2}R_{rad}=(2.5\times10^{-6})^{2}\times3.5\times10^{-2}=2.19\times10^{-13}(\mathrm{W})$$

8-3　设电偶极子天线沿东西方向放置，在远方有一移动接收台停在正南方而收到最大电场强度。在电台一单元电偶极子天线为中心的圆周在地面移动时，电场强度渐渐减小，问当电场强度减少到最大值得$1/\sqrt{2}$时，电台的位置偏离正南多少度？

解：　$f=1\mathrm{MHz}$：　$R_{\mathrm{rad}}=80\pi^{2}\left(\dfrac{\Delta l}{\lambda}\right)^{2}=80\pi^{2}\left(\dfrac{20}{300}\right)^{2}=3.5\ (\Omega)$

$$P=R_{\mathrm{rad}}I^{2}=80\pi^{2}\left(\frac{\Delta l}{\lambda}\right)^{2}I^{2}=21.9\times10^{-10}\ (\mathrm{W})$$

$f=100\mathrm{kHz}$：$R_{\mathrm{rad}}=80\pi^{2}\left(\dfrac{\Delta l}{\lambda}\right)^{2}=80\pi^{2}\left(\dfrac{20}{3\times10^{3}}\right)^{2}=3.5\times10^{-2}\ (\Omega)$

$$P=R_{\mathrm{rad}}I^{2}=3.5\times10^{-2}\times\left(2.5\times10^{-6}\right)^{2}=21.9\times10^{-14}\,(\mathrm{W})$$

8-4 自由空间中半波天线侧面 15 公里处电场强度的幅值为 0.1V/m .若工作频率为 100MHz，决定天线长度和总的辐射功率。同时写出电场强度和磁场强度的瞬时表达式。

解：$f=100\mathrm{MHz}$，$\omega=2\pi f=628.319\times10^{6}$ rad/s

$$\beta=\frac{\omega}{c}=\frac{628.319\times10^{6}}{3\times10^{8}}=2.094\ \mathrm{rad/m}$$

$$\lambda=\frac{2\pi}{\beta}=\frac{2\pi}{2.094}=3\,\mathrm{m}$$

因此半波振子天线的长度为 1.5m。由（8-5-7），当$\theta=90^{\mathrm{o}}$及$r=15\,\mathrm{km}$，电流最大值为

$$I_0=\frac{r\sin\theta}{60\cos\left(\frac{\pi}{2}\cos\theta\right)}\left|E_\theta\right|=\frac{15\times10^3}{60}\cdot0.1=25\mathrm{A}$$

半波振子天线的辐射电阻为$R_{\mathrm{rad}}=73.14\Omega$，因此天线的辐射功率为

$$P=\frac{1}{2}R_{\mathrm{rad}}I_0^2=0.5\times25^2\times73.14=22.86\ (\mathrm{kW})$$

$$E_\theta(r,\theta,\phi,t)=-\frac{1500}{r}\ \frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\sin\left(6.283\times10^8t-2.094\,r\right)\mathrm{V/m}$$

$$H_\phi(r,\theta,\phi,t)=-\frac{3.98}{r}\ \frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}\sin\left(6.283\times10^8t-2.094\,r\right)\mathrm{A/m}$$

8-5 求半波天线的主瓣宽度。

解：$\begin{cases}\dfrac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}=0.707\\ \cos^2\theta+\sin^2\theta=1\end{cases}$ 因此解得$\theta=51^\circ$

$$\Delta\theta=2\cdot(90^\circ-\theta)=78^\circ$$

8-6 一发射天线位于坐标原点，离天线较远处测得天线激发的电磁波的场强为

$$\vec{E}(r,t)=E_0\frac{\sin\theta}{r}\sin\left[\omega(t-\frac{r}{c})\right]\vec{e}_\theta\ V/m$$

式中 c 为真空中的光速。求天线辐射的平均功率。

解：根据场量与波阻抗的关系，可得磁场强度为

$$H_\phi=\frac{E_\theta}{Z_0}=\frac{E_0\sin\theta}{Z_0r}\sin\omega(t-\frac{r}{c})$$

平均功率为

$$P = R_e\left[\frac{1}{2}\oint_S\left(\dot{\vec{E}}_m \times \dot{\vec{H}}_m^*\right)\bullet d\vec{S}\right] = \frac{1}{2}\oint_S R_e \frac{E_0\sin\theta}{r}e^{j\beta r}(\vec{e_\theta}\times\vec{e_\phi})\frac{E_0\sin\theta}{Z_0 r}e^{j\beta r}\bullet d\vec{S}$$

$$= \int_0^{2\pi}\int_0^{\pi}\frac{E_0^2\sin^2\theta}{2Z_0 r^2}r^2\sin\theta d\theta d\varphi = \frac{E_0^2}{90}$$

8-7　在二元天线阵中，设 $d=\frac{\lambda}{4},\alpha=90^\circ$ ，求阵因子

解：

$$F_{12}(\theta,\phi) = \left|1+me^{i\psi}\right| = \sqrt{1+m^2+2m\cos(\beta d\sin\theta\cos\phi-\alpha)}$$

带入α=90°，$d=\frac{\lambda}{4},\beta=\frac{2\pi}{\lambda}$

$$F_{12}(\theta,\phi) = \left|1+me^{i\phi}\right| = \sqrt{1+m^2+2m\cos(\frac{\pi}{2}+\frac{\pi}{2}\sin\theta\cos\phi)}$$

8-8 两个半波天线平行放置，相距 ，他们电流的振幅相等，同激励。试用方向图乘法草绘处三个主平面的方向图

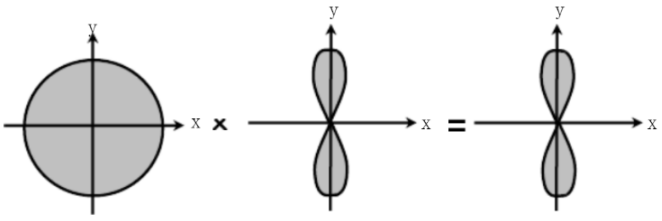
解：$d=\frac{\lambda}{2},\beta=\frac{2\pi}{\lambda},\alpha=0^\circ$，

$F_1=\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}$，令$\psi=\alpha+\beta d\sin\theta\cos\phi=\alpha+\pi\sin\theta\cos\phi$

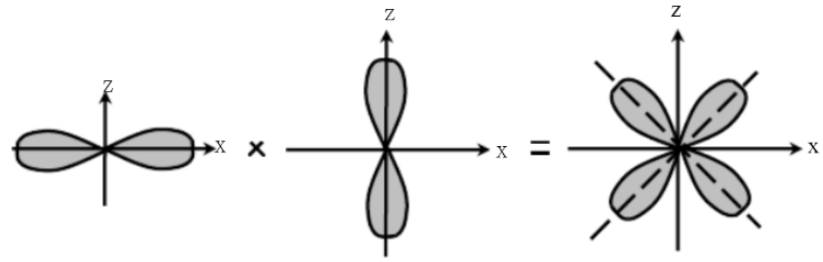
$$F_{12}=\sqrt{1+m^2+2m\cos(\alpha+\beta d\sin\theta\cos\phi)}$$

$$F(\theta,\phi)=F_1\cdot F_{12}=\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\cdot 2\cos(\frac{\pi}{2}\sin\theta\cos\phi)$$

1)　X-Y 平面：$\theta=\frac{\pi}{2}$，$F(\theta,\phi)=1\cdot 2\cos(\frac{\pi}{2}\cos\phi)$

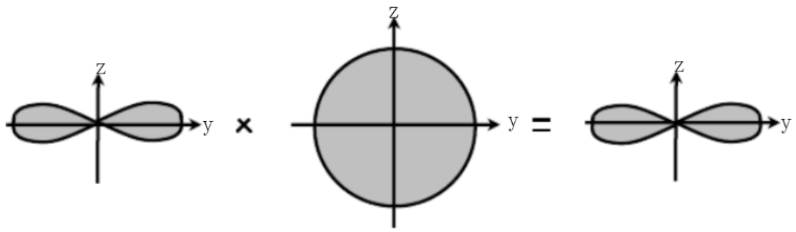


2) X-Z 平面：$\phi = 0$，$F(\theta,\phi) = \dfrac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta} \cdot 2\cos(\frac{\pi}{2}\sin\theta)$



$\phi = 0(x \circ z)$ 平面

3) Z-Y 平面：$\phi = \dfrac{\pi}{2}$，$F(\theta,\phi) = F_1 \cdot F_{12} = \dfrac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta} \cdot 2$



$\phi = \dfrac{\pi}{2}(y \circ z)$ 平面